

9.2 SUPERPOSITION THEOREM

The current through, or voltage across, an element in a linear bilateral network is equal to the algebraic sum of the currents or voltages produced independently by each source.

EXAMPLE 9.1 Determine I_1 for the network of Fig. 9.4.

Solution: Setting $E = 0$ V for the network of Fig. 9.4 results in the network of Fig. 9.5(a), where a short-circuit equivalent has replaced the 30-V source.

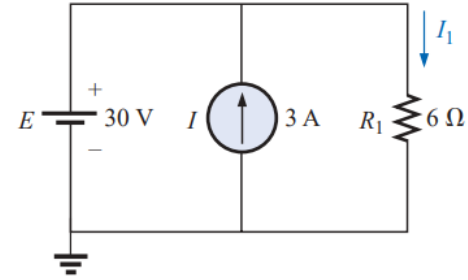


FIG. 9.4
Example 9.1.

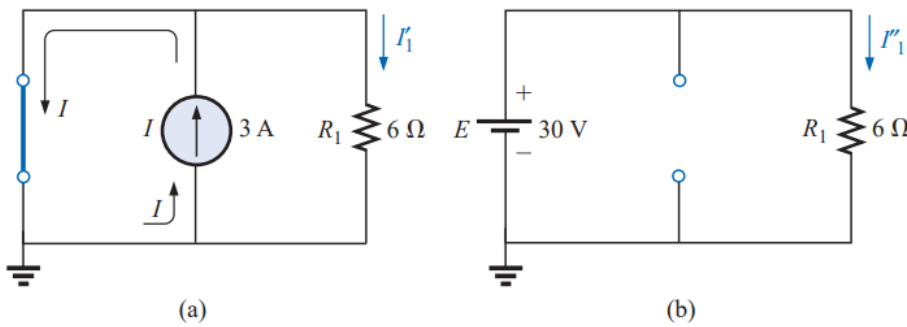


FIG. 9.5

(a) The contribution of I to I_1 ; (b) the contribution of E to I_1 .

As shown in Fig. 9.5(a), the source current will choose the short-circuit path, and $I'_1 = 0$ A. If we applied the current divider rule,

$$I'_1 = \frac{R_{sc}I}{R_{sc} + R_1} = \frac{(0 \Omega)I}{0 \Omega + 6 \Omega} = 0 \text{ A}$$

Setting I to zero amperes will result in the network of Fig. 9.5(b), with the current source replaced by an open circuit. Applying Ohm's law,

$$I''_1 = \frac{E}{R_1} = \frac{30 \text{ V}}{6 \Omega} = 5 \text{ A}$$

Since I'_1 and I''_1 have the same defined direction in Fig. 9.5(a) and (b), the current I_1 is the sum of the two, and

$$I_1 = I'_1 + I''_1 = 0 \text{ A} + 5 \text{ A} = 5 \text{ A}$$

Note in this case that the current source has no effect on the current through the 6-Ω resistor. The voltage across the resistor must be fixed at 30 V because they are parallel elements.

9.3 THÉVENIN'S THEOREM

Thévenin's theorem states the following:

Any two-terminal, linear bilateral dc network can be replaced by an equivalent circuit consisting of a voltage source and a series resistor, as shown in Fig. 9.24.

The following sequence of steps will lead to the proper value of R_{Th} and E_{Th} .

Preliminary:

1. Remove that portion of the network across which the Thévenin equivalent circuit is to be found. In Fig. 9.26(a), this requires that the load resistor R_L be temporarily removed from the network.
2. Mark the terminals of the remaining two-terminal network. (The importance of this step will become obvious as we progress through some complex networks.)

R_{Th} :

3. Calculate R_{Th} by first setting all sources to zero (voltage sources are replaced by short circuits, and current sources by open circuits) and then finding the resultant resistance between the two marked terminals. (If the internal resistance of the voltage and/or current sources is included in the original network, it must remain when the sources are set to zero.)

E_{Th} :

4. Calculate E_{Th} by first returning all sources to their original position and finding the open-circuit voltage between the marked terminals. (This step is invariably the one that will lead to the most confusion and errors. In all cases, keep in mind that it is the open-circuit potential between the two terminals marked in step 2.)

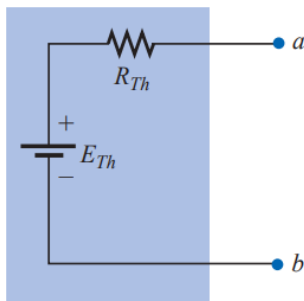


FIG. 9.24

Thévenin equivalent circuit.

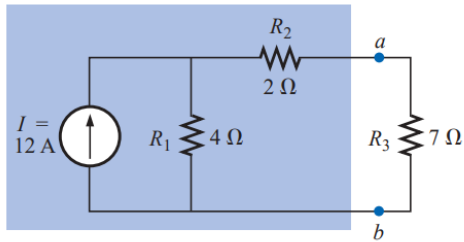


FIG. 9.33
Example 9.7.

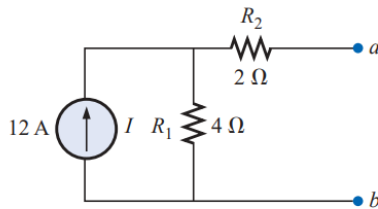


FIG. 9.34
Establishing the terminals of particular interest for the network of Fig. 9.33.

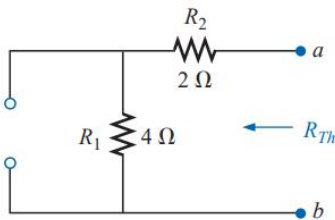


FIG. 9.35
Determining R_{Th} for the network of Fig. 9.34.

EXAMPLE 9.7 Find the Thévenin equivalent circuit for the network in the shaded area of the network of Fig. 9.33.

Solution:

Steps 1 and 2 are shown in Fig. 9.34.

Step 3 is shown in Fig. 9.35. The current source has been replaced with an open-circuit equivalent, and the resistance determined between terminals a and b .

In this case an ohmmeter connected between terminals a and b would send out a sensing current that would flow directly through R_1 and R_2 (at the same level). The result is that R_1 and R_2 are in series and the Thévenin resistance is the sum of the two.

$$R_{Th} = R_1 + R_2 = 4\ \Omega + 2\ \Omega = 6\ \Omega$$

Step 4 (Fig. 9.36): In this case, since an open circuit exists between the two marked terminals, the current is zero between these terminals and through the 2- Ω resistor. The voltage drop across R_2 is, therefore,

$$V_2 = I_2 R_2 = (0) R_2 = 0\ \text{V}$$

and
$$E_{Th} = V_1 = I_1 R_1 = I R_1 = (12\ \text{A})(4\ \Omega) = 48\ \text{V}$$

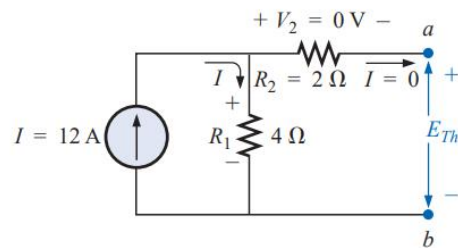


FIG. 9.36
Determining E_{Th} for the network of Fig. 9.34.

Step 5 is shown in Fig. 9.37.

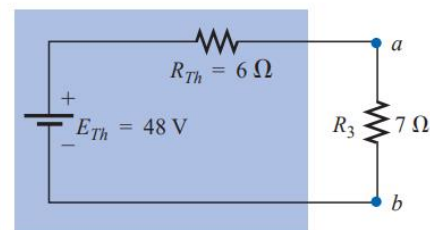


FIG. 9.37
Substituting the Thévenin equivalent circuit in the network external to the resistor R_3 of Fig. 9.33.

9.4 NORTON'S THEOREM

Any two-terminal linear bilateral dc network can be replaced by an equivalent circuit consisting of a current source and a parallel resistor, as shown in Fig. 9.58.

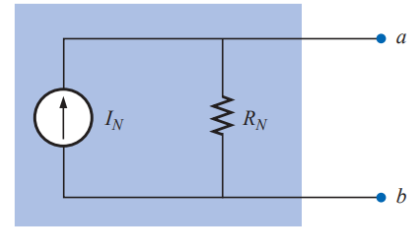


FIG. 9.58
Norton equivalent circuit.

Preliminary:

1. *Remove that portion of the network across which the Norton equivalent circuit is found.*
2. *Mark the terminals of the remaining two-terminal network.*

R_N :

3. *Calculate R_N by first setting all sources to zero (voltage sources are replaced with short circuits, and current sources with open circuits) and then finding the resultant resistance between the two marked terminals. (If the internal resistance of the voltage and/or current sources is included in the original network, it must remain when the sources are set to zero.) Since $R_N = R_{Th}$, the procedure and value obtained using the approach described for Thévenin's theorem will determine the proper value of R_N .*

I_N :

4. *Calculate I_N by first returning all sources to their original position and then finding the short-circuit current between the marked terminals. It is the same current that would be measured by an ammeter placed between the marked terminals.*

The Norton and Thévenin equivalent circuits can also be found from each other by using the source transformation discussed earlier in this chapter and reproduced in Fig. 9.59.

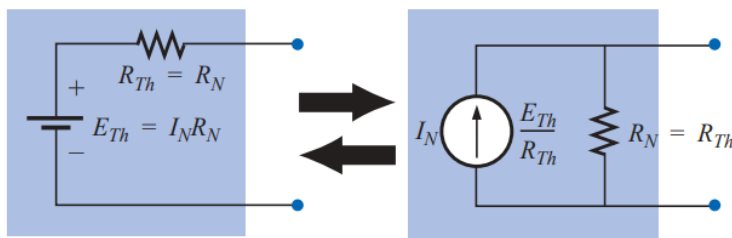


FIG. 9.59

Converting between Thévenin and Norton equivalent circuits.

EXAMPLE 9.11 Find the Norton equivalent circuit for the network in the shaded area of Fig. 9.60.

Solution:

Steps 1 and 2 are shown in Fig. 9.61.

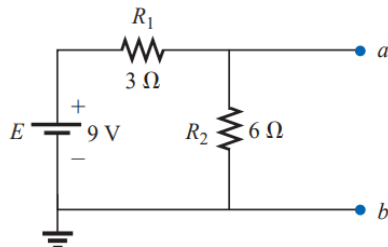


FIG. 9.61

Identifying the terminals of particular interest for the network of Fig. 9.60.

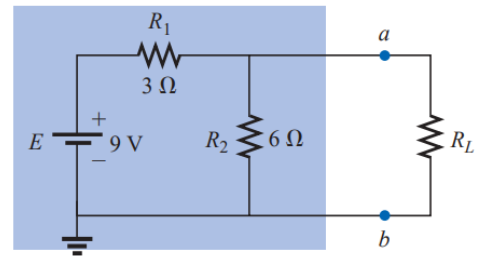


FIG. 9.60
Example 9.11.

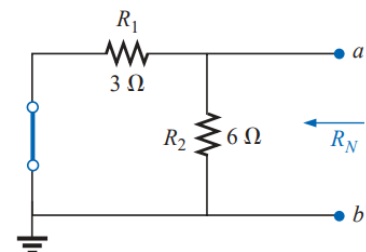


FIG. 9.62

Determining R_N for the network of Fig. 9.61.

Step 3 is shown in Fig. 9.62, and

$$R_N = R_1 \parallel R_2 = 3\ \Omega \parallel 6\ \Omega = \frac{(3\ \Omega)(6\ \Omega)}{3\ \Omega + 6\ \Omega} = \frac{18\ \Omega}{9} = 2\ \Omega$$

Step 4 is shown in Fig. 9.63, clearly indicating that the short-circuit connection between terminals a and b is in parallel with R_2 and eliminates its effect. I_N is therefore the same as through R_1 , and the full battery voltage appears across R_1 since

$$V_2 = I_2 R_2 = (0)6\ \Omega = 0\ \text{V}$$

Therefore,

$$I_N = \frac{E}{R_1} = \frac{9\ \text{V}}{3\ \Omega} = 3\ \text{A}$$

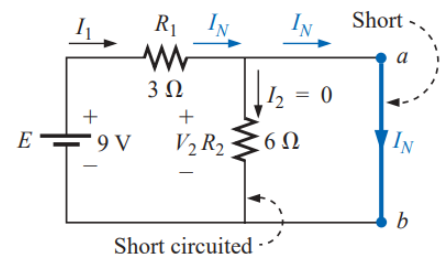


FIG. 9.63

Determining I_N for the network of Fig. 9.61.

Step 5: See Fig. 9.64. This circuit is the same as the first one considered in the development of Thévenin's theorem. A simple conversion indicates that the Thévenin circuits are, in fact, the same (Fig. 9.65).

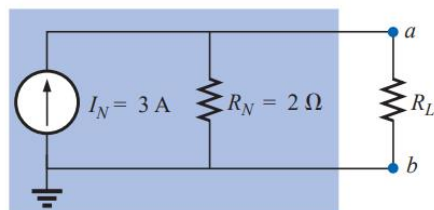


FIG. 9.64

Substituting the Norton equivalent circuit for the network external to the resistor R_L of Fig. 9.60.

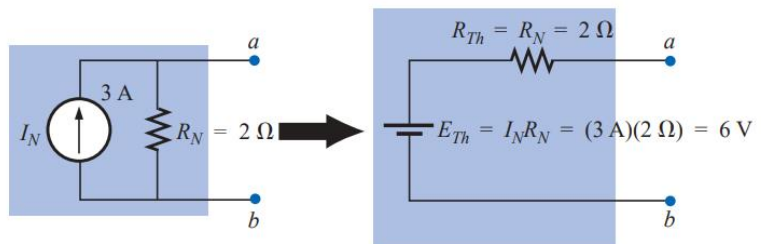


FIG. 9.65

Converting the Norton equivalent circuit of Fig. 9.64 to a Thévenin equivalent circuit.

Maximum Power Transfer

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).

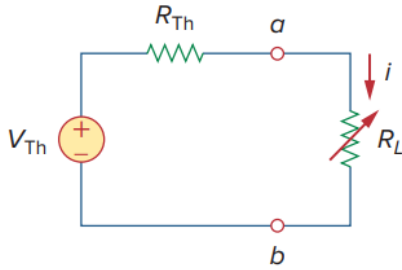


Figure 4.48

The circuit used for maximum power transfer.

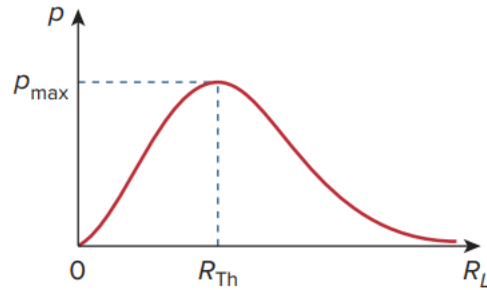


Figure 4.49

Power delivered to the load as a function of R_L .

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

To prove the maximum power transfer theorem, we differentiate p in Eq. (4.21) with respect to R_L and set the result equal to zero. We obtain

$$\begin{aligned} \frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0 \end{aligned}$$

This implies that

$$0 = (R_{Th} + R_L - 2R_L) = (R_{Th} - R_L)$$

which yields

$$R_L = R_{Th}$$

$$p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$$